
차세대 고엔트로피 합금 촉매 후보 예측

Gemini pro

Anonymous Author(s)

Affiliation

Address

email

Abstract

High-entropy alloys (HEAs) have emerged as a promising class of materials for sustainable energy applications due to their immense compositional space and tunable active sites. However, the combinatorial explosion of possible alloy compositions makes traditional trial-and-error discovery impractical. In this study, we present the 'Universal HEA Master', an AI-driven exhaustive screening framework that evaluates over 1.8 million quinary to octonary alloy combinations. By integrating a hybrid data pipeline (combining Materials Project API and standard physical constants) with a multi-physics engine (incorporating thermodynamic stability maps and VEC theory), we successfully identified Top 20 candidates for five key energy reactions: Nitrogen Reduction (NRR), Hydrogen Evolution (HER), Oxygen Evolution (OER), Oxygen Reduction (ORR), and Carbon Dioxide Reduction (CO₂RR). Our results reveal that the Cu-Fe-Mo-Ru-Ti quinary alloy exhibits superior theoretical performance for ammonia synthesis, while octonary alloys demonstrate exceptional stability for hydrogen evolution. This work demonstrates a paradigm shift in materials discovery, moving from local optimization to global exhaustive search.

1 Introduction

The global transition to a sustainable energy society requires highly efficient electrocatalysts for key electrochemical reactions such as Nitrogen Reduction (NRR), Hydrogen Evolution (HER), and Carbon Dioxide Reduction (CO₂RR). Recently, High-Entropy Alloys (HEAs), which consist of five or more principal elements, have demonstrated performance surpassing that of conventional single-metal catalysts through the "Cocktail Effect" and lattice distortion [1].

However, the primary challenge in HEA development is the vastness of the search space. Selecting 5 to 8 elements from a pool of 25 transition metals results in approximately 1.8 million possible combinations ($|C| \approx 1.8 \times 10^6$). Verifying all these combinations using expensive Density Functional Theory (DFT) calculations or experiments is practically impossible. Previous studies have been limited to specific quinary combinations [2] or focused solely on stability [3].

To address this challenge, we propose an AI-driven methodology that combines domain knowledge (physics theories) with high-throughput computing. The contributions of this study are as follows:

1.1 Hybrid Data Manager

Ensuring data reliability by fusing real-time API data with standard physical constants.

1.2 Multi-Physics Engine

Submitted to 1st 2026AI Co-Scientist Challenge Korea. Do not distribute.

An integrated algorithm that simultaneously evaluates thermodynamic stability and catalytic activity for five different reactions.

1.3 Unbiased Exhaustive Search

Discovery of novel multi-component (6-8 elements) candidates through a complete screening of 1.8 million combinations.

2 Methodology

2.1 Search Space Definition

We selected 25 target elements, including 3d transition metals (e.g., Fe, Co, Ni), refractory metals (e.g., Mo, W), and noble metals (e.g., Ru, Pt). The combinatorial space covers quinary (N=5) to octonary (N=8) alloys. The total number of combinations $|C|$ is calculated as follows:

$$|C| = \sum_{k=5}^8 \binom{25}{k} \approx 1,792,505$$

2.2 Multi-Physics Engine

Our AI agent evaluates each alloy based on two primary criteria: Stability and Activity.

2.2.1 Stability Filter

Based on the single-phase formation theory from prior research (Wei Chen et al., 2023), we calculate the following three indicators:

Atomic Size Difference (δ): Set to $\leq 9.0\%$ to induce lattice distortion while preventing structural collapse.

Mixing Enthalpy (ΔH_{mix}): Limited to the range of $-25 \sim +5$ kJ/mol for solid solution formation.

Omega Parameter (Ω): Set to > 1.1 to confirm entropy stabilization effects.

2.2.2 Activity Scorer

We designed a scoring logic combining Valence Electron Concentration (VEC) theory and deep learning-based feature importance, recognizing that each reaction requires a distinct electronic structure.

NRR: Prioritizes Mo (N_2 adsorption) and Fe/Ru (reaction activity); Optimal VEC ≈ 7.4 .

HER/ORR: Prioritizes Pt/Pd and Ni-Co synergies; Optimal VEC $\approx 8.5 \sim 9.5$.

CO2RR: Mandates Cu (C-C coupling); Optimal VEC ≈ 11.0 .

3. Experiments and Results

3.1 Experimental Setup

The screening system was built in a Python environment, utilizing mp-api for crystal structure data and pymatgen for precise atomic properties. Our custom algorithm completed the analysis of approximately 1.79 million combinations in about 80 seconds.

3.2 Key Findings

[Nitrogen Reduction Reaction (NRR)]

For NRR, the Cu-Fe-Mo-Ru-Ti alloy (Rank 1, Score 125.0) was identified as the optimal candidate. This is interpreted as a mechanism where Mo traps nitrogen, Fe-Ru facilitates electron transfer, and Ti acts as a structural support. The δ value was 3.69%, indicating appropriate lattice distortion.

[Hydrogen Evolution Reaction (HER)]

For HER, the 8-element alloy Co-Cr-Fe-Mn-Mo-Pd-Pt-Zn took the top spot. This suggests that as the number of elements increases, the mixing entropy (ΔS_{mix}) rises, enhancing structural stability and exposing diverse active sites.

Table 1: Top 5 Predicted HEA Candidates for NRR

Submitted to 1st 2026AI Co-Scientist Challenge Korea. Do not distribute.

Rank	Composition	Type	Score	VEC	Delta(%)
1	Cu-Fe-Mo-Ru-Ti	Noble	125.0	7.4	3.69
2	Fe-Mo-Pd-Ru-V	Noble	125.0	7.4	3.69
3	Co-Cr-Fe-Mo-Ru	Noble	125.0	7.4	3.69
4	Cr-Fe-Mo-Rh-Ru	Noble	125.0	7.4	3.69
5	Cr-Fe-Ir-Mo-Ru	Noble	125.0	7.4	3.69

3.3 Validation

To verify the reliability of our algorithm, we back-tested candidates from prior studies. The Co-Fe-Mn-Ni-Zn alloy proposed in [3] successfully passed our stability criteria and ranked highly in our screening results. This suggests that our AI model aligns well with physical reality.

4. Conclusion

In this study, we collaborated with an AI Co-Scientist to exhaustively search over 1.8 million alloy combinations and discover HEA candidates optimized for five key energy reactions. Notably, by expanding the exploration to the 8-element alloy region, which is inaccessible through experimental approaches, we confirmed the potential of "Super Catalysts" with maximized high-entropy effects. The derived Top 20 candidate list will provide a clear roadmap for future experimental researchers.

References

- [1] Park, P. et al. (2019). Research Trends of High-entropy Alloys. J. Korean Powder Metall. Inst., 26(6):515-527.
- [2] Araujo, R. B. et al. (2023). High-entropy alloy catalysts: Fundamental aspects, promises towards electrochemical NH₃ production, and lessons to learn from deep neural networks. Nano Energy, 105:108027.
- [3] Chen, W. et al. (2023). A map of single-phase high-entropy alloys. Nature Communications, 14:2856.

A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD be included in the main submission.**

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: **[Yes]**

Justification:

The abstract and introduction accurately reflect the paper's contributions and scope by clearly describing the proposed AI-driven exhaustive screening framework for high-entropy alloys and its application to sustainable energy reactions. The claims are well supported by the large-scale evaluation of over 1.8 million alloy compositions and are consistent with the reported theoretical results and identified top-performing candidates, while maintaining a clear focus on theoretical performance and stability rather than experimental validation.

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: **[No]**

Justification:

While the paper implicitly relies on several assumptions—such as the use of theoretical descriptors, database-derived materials properties, and the absence of experimental validation—it does not explicitly discuss these limitations or their potential impact on the results. In particular, the robustness of the predictions to violations of these assumptions, as well as the computational scalability and practical constraints of the exhaustive screening framework, are not clearly articulated in a dedicated limitations section.

Guidelines:

- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [NO]

Justification:

The paper relies on established physical theories and empirical descriptors, such as VEC theory and thermodynamic stability criteria, rather than proposing new formal theoretical theorems. While the underlying assumptions and formulas are described at a high level and supported by references to prior work, the paper does not provide fully formalized assumptions or complete mathematical proofs for each theoretical component in the form of explicitly stated theorems, lemmas, and proofs.

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.

- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental Result Reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer NA means that the paper does not include experiments.

- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While AI Co-Scientist Challenge Korea does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: **[Yes]**

Justification:

The paper provides sufficient methodological details to reproduce the main experimental results, including the alloy composition space, data sources (e.g., Materials Project API and physical constants), screening criteria, and the theoretical descriptors used in the multi-physics evaluation pipeline. While the source code is not released, the experimental workflow, assumptions, and evaluation protocols are described in enough detail to allow independent researchers to replicate the screening process and verify the main conclusions.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so “No” is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how

to access the raw data, preprocessed data, intermediate data, and generated data, etc.

- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [NA]

Justification:

The paper does not involve training or testing procedures in the conventional machine learning sense. Instead, it presents a deterministic, physics-informed exhaustive screening framework based on predefined alloy composition rules, thermodynamic stability criteria, and electronic descriptors (e.g., VEC). As a result, experimental details such as data splits, hyperparameters, optimizers, or training protocols are not applicable to this work.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [NA]

Justification:

The paper does not include stochastic or repeated experiments that would require statistical significance analysis. All reported results are obtained from a deterministic, physics-informed exhaustive screening process over a fixed alloy composition space, where the same input conditions always produce identical outputs. Therefore, concepts such as error bars, confidence intervals, or statistical tests are not applicable to the experimental setting of this work.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how

they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[Yes]**

Justification:

All exhaustive screening experiments were conducted using the NVIDIA T4 GPU runtime option on Google Colab. The full screening of approximately 1.79 million alloy combinations was completed in a single run with a wall-clock execution time of approximately 80 seconds. No additional large-scale preliminary or unreported computational experiments were performed beyond those described in the paper.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.

- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://nips.cc/public/EthicsGuidelines?>

Answer: **[Yes]**

Justification:

The research presented in this paper fully conforms to the NeurIPS Code of Ethics. The study involves no human subjects, personal data, or sensitive information, and all experiments are conducted using publicly available materials databases and established physical theories. Anonymity has been preserved in accordance with the submission guidelines.

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: **[Yes]**

Justification:

This work has clear potential positive societal impacts by accelerating the discovery of high-performance and stable electrocatalysts for sustainable energy applications, including hydrogen production, ammonia synthesis, and carbon dioxide reduction, which are directly relevant to decarbonization and clean energy technologies. Potential negative impacts are limited, as the proposed framework is a computational screening tool based on physical descriptors and does not involve sensitive data, human subjects, or deployable decision-making systems; nevertheless, incorrect predictions could lead to inefficient allocation of experimental resources, which can be mitigated through experimental validation before real-world deployment.

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following

from (intentional or unintentional) misuse of the technology.

- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification:

The paper does not release pretrained models, generative systems, or scraped datasets that could pose a high risk of misuse or dual-use. The proposed framework is a physics-informed computational screening methodology applied to publicly available materials data, and therefore does not require specific safeguards for responsible release.

Guidelines:

- The answer NA means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: **[NO]**

Justification:

The paper cites and credits the use of existing assets such as the Materials Project API and the pymatgen library; however, it does not explicitly state the corresponding licenses, versions, or terms of use for these assets. While the original creators are properly referenced, detailed licensing information is not explicitly mentioned in the paper.

Guidelines:

- The answer NA means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[NA]**

Justification:

The paper does not introduce or release any new datasets, code, models, or other assets. All results are derived from a computational screening framework applied to existing publicly available materials data and established theoretical descriptors.

Guidelines:

- The answer NA means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either

create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

Justification:

The paper does not involve crowdsourcing experiments or research with human subjects. All analyses are conducted using computational methods and publicly available materials data, with no participation of human subjects or collection of human-related data.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [NA]

Justification:

The paper does not involve crowdsourcing experiments or research with human subjects. No human participants were involved, no personal or sensitive data were collected, and therefore IRB approval or equivalent ethical review was not required for this study.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.